Dynamic control of light scattering in a single particle enabled by time modulation

Mohammad Mojtaba Sadafi,[1] Achiles Fontana da Mota,[1,2] and Hossein Mosallaei[1,a)]

[1)] *Metamaterials Laboratory, Electrical and Computer Engineering Department, Northeastern University, Boston, MA 02115, USA*

[2)] *Department of Electrical Engineering, University of Brasilia, Brasilia, DF 70910-900, Brazil*

The interaction of light with time-varying materials offers intriguing opportunities for controlling light scattering and wavefront manipulation, thereby unlocking fascinating applications in the realm of optics and photonics. In this study, we present an analytical solution for the scattering from a particle made of a material with time-varying permittivity by exploiting the T-matrix approach. Through the manipulation of the active medium's eigenvalues, we demonstrate the pivotal ability to regulate the elements of a dynamically controlled T-matrix, thus enabling precise control over the scattering characteristics of the particle. Crucially, this dynamic control is achieved without resorting to modifying the particle's inherent physical parameters, such as shape, size, dispersion, etc. We demonstrate that the eigenvalues of the dynamic material can be skillfully manipulated through the adequate choice of the particle's modulation function, resulting in either in-phase or out-of-phase interactions between the magnetic and electric dipole modes, allowing us to satisfy the Kerker conditions at diverse harmonics. The results of the optimal modulation functions are presented in both the near-field and far-field regions, revealing time modulation as a dynamic means of achieving unidirectional scattering. Our findings pave the way for developing time-varying structures comprising dynamic meta-atoms, offering valuable insights into advanced light-matter interaction and providing lucrative guidance for future research in the realm of dynamic photonic systems.

[a)] hosseinm@coe.neu.edu

Investigating light interaction with structures comprising time-varying materials has sparked tremendous interest as they introduce an additional degree of freedom to the design paradigm[2–5]. Ranging from microwaves to optics, the utilization of time-varying materials has opened up thrilling possibilities for diverse applications, including beam steering[6–10], nonreciprocity[11–14], amplification[15–17], exceptional points[18–20], and signal processing[21–23]. However, realizing such applications within the infrared and visible range necessitates materials with tailored optical characteristics to facilitate the desired variations[29,30]. In order to achieve this objective, several approaches have been explored, including free carrier density modulation[24,25], acousto-optic[26,27], and magneto-optic[28] techniques, enabling favorable variations in the optical characteristics of materials.

The majority of the proposed structures presented in the literature are predominantly situated within unbounded media and flat surfaces, encompassing intriguing components like slabs with time-varying permittivity[31], temporal photonic crystals[32–34], and time-modulated metasurfaces[35–37]. Although extensive efforts have been devoted to exploring time modulation in infinite media, the realm of light-matter interaction with finite time-varying structures is a burgeoning domain that has only been addressed in a limited number of works. Specifically, the scattering of light from a spherical particle composed of a material exhibiting periodic permittivity variations over time has been meticulously investigated[38], leading the authors to reveal an intricate energy exchange mechanism between the incident light and the dynamic medium. Furthermore, in[39], a comprehensive exploration into the scattering properties of a time-varying spherical conducting shell has been conducted, employing the T-matrix approach. Additionally, comprehensive studies concerning the scattering of light from a spherical particle constituted of a dispersive material modulated in time have been undertaken, uncovering the profound capability, subject to the prudent selection of the incident field, to transfer energy from the time modulation to the electromagnetic field[40]. Moreover, through rigorous investigations, it has been demonstrated that time modulation in a particle made of a material with proper Lorentzian dispersion can facilitate parametric amplification, followed by the emergence of desired radiation patterns contingent upon the appropriate selection of geometric parameters of the scatterer[41].

The phenomenon of light scattering from a homogeneous sphere, commonly referred to as the Lorenz-Mie theory, has been extensively studied and well-documented in numerous prior investigations[42–46]. By expressing the electromagnetic fields in terms of multipoles,

a profound understanding of the underlying physics governing the scattering characteristics of particles emerges, presenting opportunities to design more intricate and sophisticated devices[47–49]. More specifically, the strategic design of spherical particles with electric and magnetic dipole modes operating at the same frequency has facilitated their utility as building blocks in double negative (DNG) metamaterials, thereby enabling thrilling applications[47]. Additionally, precise control over the directionality of light scattering is achieved by manipulating the interference between electric and magnetic dipoles within an asymmetric dimer of silicon nanoparticles[48]. Thoroughly engineered electric and magnetic dipole modes are spectrally aligned and exhibit equal strengths, empowering the realization of unidirectional light scattering[49]. Notwithstanding these remarkable achievements, the aforementioned approaches predominantly rely on tuning the geometric properties of the particle to attain the desired functionalities, and the design of more intricate devices necessitates the incorporation of strong spatial dispersion effects[50–53].

In this manuscript, an additional degree of freedom is proposed by a time-modulated scatterer, besides its inherent geometrical properties, that can be tuned to exhibit fascinating scattering characteristics. It is demonstrated how time modulation can enable the simultaneous regulation of the magnitude and phase of the excited multipole moments in the particle, all without reliance on its physical parameters such as shape, size, dispersion, etc. This consequently leads to the dynamic control of light scattered from the particle. An exhaustive analytical platform is introduced to derive the Mie coefficients of a time-modulated spherical scatterer. We reveal how different multipolar modes, excited within the particle, can be frequency converted and engineered through time variation, resulting in real-time light control. Particularly, we show how judicious selection of the modulation function, using particle swarm optimization techniques, enables the attainment of a tunable in-phase and out-of-phase interference of the electric and magnetic dipoles. This phenomenon, also known as the Kerker conditions[54], facilitates the generation of uni-directional light scattering from the particle.

The scattering problem is tackled through the utilization of the time-Floquet approach, building upon prior works[38]. Consideration is given to a spherical particle characterized by a radius, R, and possessing a periodic time-varying permittivity denoted as $\epsilon_r(t) = \epsilon_r(t+T_0)$, with T_0 representing the period. The particle is situated within a medium characterized by permittivity ϵ_m. The permittivity function, $\epsilon_r(t)$, can be suitably expanded in terms of a

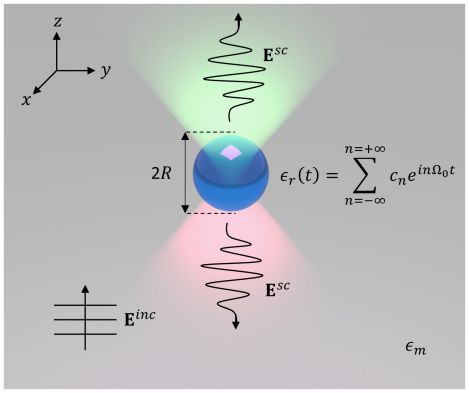


FIG. 1: Schematic depiction of a time-varying scatterer with the permittivity of $\epsilon_r(t)$ and radius R placed in a medium with permittivity ϵ_m.

Fourier series, as expressed below:

$$\epsilon_r(t) = \sum_{n=-\infty}^{+\infty} c_n e^{in\Omega_0 t}, \tag{1}$$

where c_n represents the Fourier coefficient of $\epsilon_r(t)$, and it adheres to the condition $c_{-n} = c_n^*$ to ensure that $\epsilon_r(t)$ remains a real-valued signal, and $\Omega_0 = \dfrac{2\pi}{T_0}$ denoting the angular frequency of modulation. The total electric field can be expressed as a combination of three distinct contributions: the incident field, the scattered field, and the internal field, defined as follows:

$$\mathbf{E}(\mathbf{r}, t) = \sum_{\iota} \mathbf{E}^{\iota}(\mathbf{r}, t). \qquad \iota \in \{inc, sc, int\} \tag{2}$$

Based on the time-Floquet theory, each component of the electric field can be expressed as a Fourier series, as shown below:

$$\mathbf{E}^{\iota}(\mathbf{r}, t) = \Re\left\{ \sum_{n=-\infty}^{+\infty} \mathbf{E}_n^{\iota}(\mathbf{r}) e^{-i(\omega_0 - n\Omega_0)t} \right\}, \tag{3}$$

in which ω_0 indicates the central angular frequency imposed by the incident light, and $\mathbf{E}_n^{\iota}(\mathbf{r})$ represents the Fourier coefficient, which can be expanded in terms of the electromagnetic multipoles. For the incident field with the central angular frequency of ω_0 and the harmonic order of p ($\omega_{inc} = \omega_0 - p\Omega_0$), the Fourier coefficient can be expressed as follows:

$$\begin{aligned} \mathbf{E}_n^{inc}(\mathbf{r}) = \sum_{\alpha} \Big[& a_{n\ l}^{inc\,m} \mathbf{M}_{l}^{rm}(\kappa_n, \mathbf{r}) + \\ & b_{n\ l}^{inc\,m} \mathbf{N}_{l}^{rm}(\kappa, \mathbf{r}) \Big] \delta_{np}, \qquad \alpha \in \{l, m\} \end{aligned} \tag{4}$$

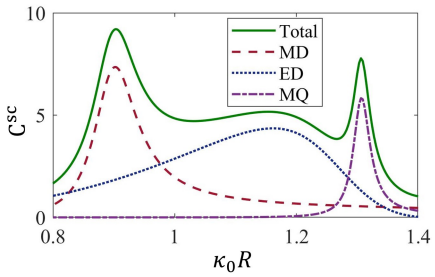


FIG. 2: Scattering cross-section of the particle with the permittivity of $\epsilon_r = 11.56$ in the absence of time modulation.

and for the scattered field is given below:

$$\mathbf{E}_n^{sc}(\mathbf{r}) = \sum_{\alpha} \left[a_{n\,l}^{scm} \mathbf{M}_l^{sm}(\kappa_n, \mathbf{r}) + b_{n\,l}^{scm} \mathbf{N}_l^{sm}(\kappa, \mathbf{r}) \right], \tag{5}$$

where the magnetic and electric multipole moments are denoted as $a_{nl}^{\iota\,m}$ and $b_{nl}^{\iota\,m}$, respectively; the regular (singular) magnetic and electric multipoles are represented by $\mathbf{M}_l^{rm}(\kappa_n, \mathbf{r})$ $(\mathbf{M}_l^{sm}(\kappa_n, \mathbf{r}))$ and $\mathbf{N}_l^{rm}(\kappa_n, \mathbf{r})$ $(\mathbf{N}_l^{rm}(\kappa_n, \mathbf{r}))$, correspondingly. Additionally, $\kappa_n = \dfrac{\omega_0 - n\Omega_0}{c}\sqrt{\epsilon_m}$ is the wavenumber associated to each harmonic in the background medium, and the parameters l and m belong to the integer sets $[1, +\infty)$ and $[-l, +l]$, respectively, representing the degree and order of the multipoles. However, the internal electric field can be effectively characterized using the eigenvalue expansion within the dynamic medium, and it can be expressed as follows:

$$\mathbf{E}_n^{int}(\mathbf{r}) = \sum_{u=-\infty}^{+\infty} \sum_{\alpha} \left[a_{u\;\;l}^{int\,m} \mathbf{u}_n^u \mathbf{M}_l^{rm}(\kappa_u, \mathbf{r}) + b_{u\;\;l}^{int\,m} \mathbf{u}_n^u \mathbf{N}_l^{rm}(\kappa_u, \mathbf{r}) \right], \tag{6}$$

in which κ_u is the eigenvalue, and $\mathbf{u}_n^u$ represents its corresponding eigenvector associated with the time-varying medium. After truncation of the summation over n to the order N_n, the dynamic medium can be described using the following eigenvalue equation:

$$\mathbf{A}\mathbf{u} = \kappa^2 \mathbf{u}, \tag{7}$$

where $\mathbf{A}$ is a square matrix of rank $2N_n + 1$ described as follows:

$$\mathbf{A} = \begin{bmatrix} c_0\gamma_{-N_n} & c_{-1} & \cdots & c_{-2N_n} \\ c_1 & c_0\gamma_{-N_n+1} & \cdots & c_{-2N_n+1} \\ \vdots & \vdots & \ddots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ c_{2N_n} & c_{2N_n-1} & \cdots & c_0\gamma_{-N_n} \end{bmatrix}, \tag{8}$$

with $\gamma_n = \dfrac{(\omega_0 - n\Omega_0)^2}{c^2}$. Therefore, having written all the fields in terms of the electromagnetic multipoles, and after truncating the summation over l to the order N_l, the unknown coefficients of the scattered and internal fields can be related to the incident field coefficients through a matrix relation as illustrated below:

$$\begin{bmatrix} \mathbf{T}_{TE} & \mathbf{0} \\ \mathbf{0} & \mathbf{T}_{TM} \end{bmatrix} \begin{bmatrix} \mathbf{a}^{inc} \\ \mathbf{b}^{inc} \end{bmatrix} = \begin{bmatrix} \mathbf{a}^{sc} \\ \mathbf{b}^{sc} \end{bmatrix}, \tag{9a}$$

$$\begin{bmatrix} \mathbf{I}_{TE} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_{TM} \end{bmatrix} \begin{bmatrix} \mathbf{a}^{inc} \\ \mathbf{b}^{inc} \end{bmatrix} = \begin{bmatrix} \mathbf{a}^{int} \\ \mathbf{b}^{int} \end{bmatrix}, \tag{9b}$$

where the entries, $T_{TE}{}_{lp}^{mn}$, $T_{TE}{}_{lp}^{mn}$, $I_{TE}{}_{lp}^{mn}$, and $I_{TM}{}_{lp}^{mn}$ establish a one-to-one mapping between the coefficients of the magnetic and electric multipoles of the incident field and those of the scattered and internal fields, respectively. These mappings are determined by enforcing the continuity of the tangential electromagnetic fields on the surface of the scatterer (a detailed description of the theoretical model and the calculations for these entries can be found in the supplementary material).

Eq. (9a) introduces the concept of a time-varying T-matrix, where the diagonal elements represent the magnetic and electric Mie coefficients of a dynamic sphere. Notably, the derived coefficients account for frequencies beyond the adiabatic approximation, where energy exchange occurs between the time modulation process and the electromagnetic field[2]. In fact, when an incoming monochromatic light interacts with the particle, it excites various harmonics, each being associated with different multipole moments. The time-varying permittivity, experimentally realizable at optical frequencies[55,56], offers the unique opportunity to simultaneously control the amplitude and phase of the excited multipole moments at each

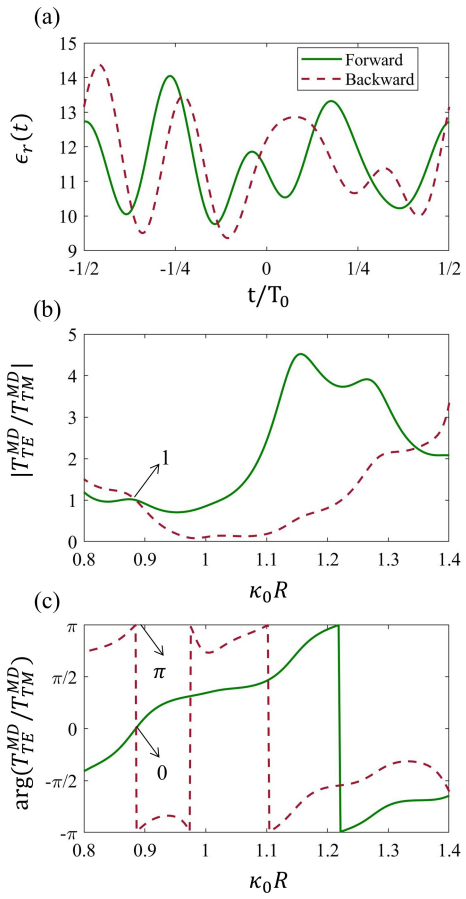


FIG. 3: The optimal modulation functions with the angular modulation frequency of $\Omega_0 = \dfrac{\omega_{MD} - \omega_{ED}}{2}$ to achieve forward and backward scattering once the particle is excited at the MD resonant frequency (a). The relative amplitude (b) and phase (b) between the dipolar magnetic and electric coefficients of the -1 harmonic order.

harmonic, leading to desirable scattering characteristics. In our study, we investigate the interaction of an x-polarized plane wave propagating in the z-direction with a particle characterized by a DC permittivity value of $\epsilon_c = 11.56$, which varies periodically in time. The frequency range under consideration is limited within a region where the particle, without any time-varying permittivity, exhibits three distinct resonances: magnetic dipole (MD), electric dipole (ED), and magnetic quadrupole (MQ), as depicted in Fig. 2. It is essential to mention that we take into account a broad range of harmonics to obtain comprehensive results, including those beyond the frequency range shown in Fig. 2. This approach allows for a thorough exploration of the particle's scattering characteristics across a wide frequency spectrum, encompassing various harmonic modes.

Following the Kerker conditions, achieving appropriate excitation of the ED and MD modes within the particle can lead to either perfect forward or backward scattering. Specifically, to suppress forward scattering, $T_{TE_{1p}}^{mn} = -T_{TM_{1p}}^{mn}$, while for backward scattering suppression, $T_{TE_{1p}}^{mn} = T_{TM_{1p}}^{mn}$. Judicious choice of the modulation function allows for the manipulation of the eigenvalues of the dynamic matter, enabling simultaneous control over the amplitude and phase of the Mie coefficients. By exploiting this characteristic, the desired forward and backward scattering conditions can be effectively satisfied. Thereafter, we engineer the permittivity by optimizing the coefficients introduced in Eq. 5 to attain the desired forward and backward scattering characteristics. Importantly, despite the phase delay between the incoming light and the modulation function, which affects the power exchanged between the light and the time modulation subsystem[57], the radiation pattern at different harmonics remains unchanged, happening on top of the fact that all the excited multipole moments experience the phase delay simultaneously. The results are obtained for both scenarios where the incoming light excites either the MD or ED in the particle since the coupling strength to the harmonics reaches its maximum under the resonant condition. In both cases, the modulation frequency is chosen as $\Omega_0 = \dfrac{\omega_{MD} - \omega_{ED}}{2}$, ensuring that the conditions are satisfied at the same frequency. Here, ω_{MD} and ω_{ED} represent the resonant frequencies of the ED and MD modes, respectively.

When the incoming field with the angular frequency ω_{MD} interacts with the particle, it not only excites the MD mode with the same angular frequency but also leads to the generation of multiple monochromatic waves known as harmonics. To achieve forward and backward scattering for the -1 harmonic, the optimal Fourier coefficients of the permittivity are determined using the genetic algorithm (GA) (for further details on the optimization procedure, please refer to the supplementary material). The resulting modulation functions obtained from the GA optimization are depicted in Fig. 3(a). Additionally, the relative amplitude and phase between the dipolar magnetic ($T_{TE}^{MD} = T_{TE}{}_{10}^{m-1}$) and electric ($T_{TM}^{MD} = T_{TM}{}_{10}^{m-1}$) Mie coefficients are illustrated in Fig. 3(b) and Fig. 3(c), respectively.

In Fig. 3(b), it is evident that the relative amplitude for both forward and backward scattering cases strictly crosses unity precisely at the resonant frequency of the MD resonance. Furthermore, in the forward scattering case, the relative phase between the magnetic and electric Mie coefficients becomes zero at the MD resonant frequency, indicating an in-phase interference between the magnetic and electric dipole modes within the scatterer. Con-

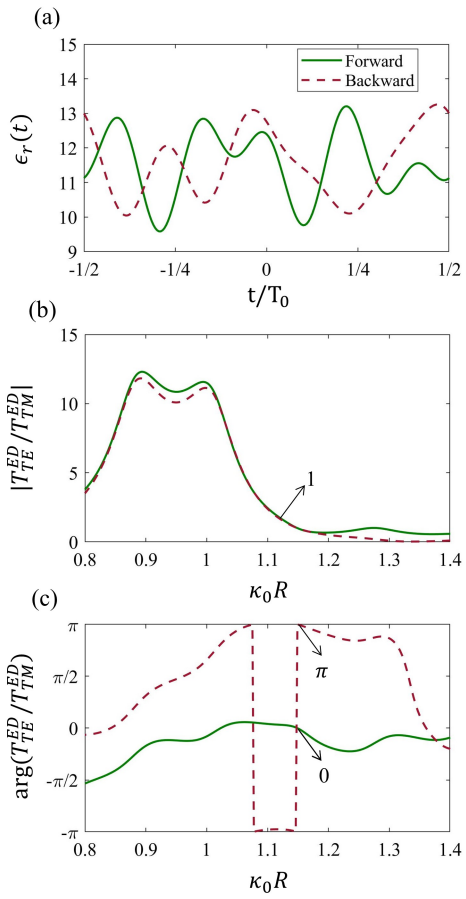


FIG. 4: The optimal solutions of the time-varying permittivity functions with the angular modulation frequency of $\Omega_0 = \dfrac{\omega_{MD} - \omega_{ED}}{2}$ to obtain forward and backward scattering for +1 harmonic when the particle is excited at the ED resonant frequency (a). The relative amplitude and phase between the +1 harmonic of magnetic and electric Mie coefficients for both forward and backward scenarios.

versely, in the backward scattering case, the excited modes have an out-of-phase interaction, resulting in a π phase difference between the coefficients at the MD resonant frequency, as depicted in Fig. 3(c). This unique combination of relations is sufficient to fulfill the Kerker conditions within the particle, leading to the realization of the desired functionalities without the need for manipulating the geometrical parameters or introducing magnetic properties, as required in the original Kerker conditions.

Moreover, following the interaction of the particle with an incoming wave associated with the angular frequency ω_{ED}, the ED mode is excited at its original angular frequency. Additionally, other monochromatic modes are stimulated due to the interaction, and their

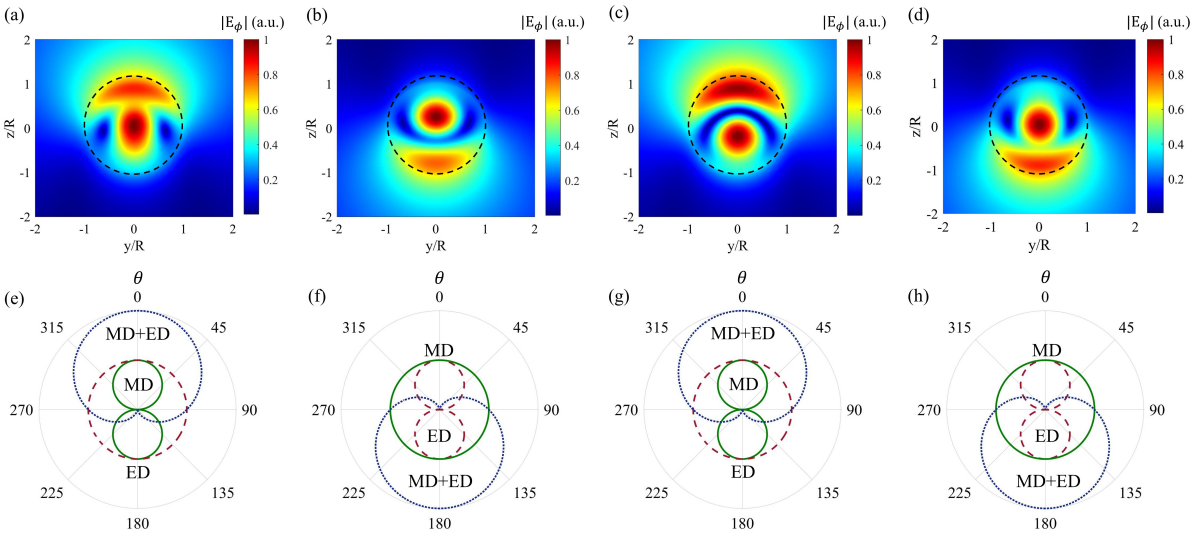


FIG. 5: In-plane components of the electric in the near-field (a-d) and far-field (e-h) regions for the forward (a,e) and backward (b,f) MD excitation and for the forward (c,g) and backward (d,h) ED excitation.

amplitudes and phases can be simultaneously controlled by utilizing the time-variation of the permittivity to achieve the desired scattering characteristics. In order to accomplish this goal, the coefficients of the permittivity are carefully selected through GA optimization to attain favorable interaction between the excited MD and ED modes within the particle. The optimal time-varying functions of the permittivity for both forward and backward scattering scenarios are demonstrated in Fig. 4(a). For each case, Fig. 4(b) and Fig. 4(c) present the relative amplitude and phase between the magnetic ($T_{TE}^{ED} = T_{TE}{}_{10}^{m+1}$) and electric ($T_{TM}^{ED} = T_{TM}{}_{10}^{m+1}$) Mie coefficients for the +1 harmonic, respectively.

As observed from Fig. 4(b), at the ED resonant frequency, the relative amplitude between the +1 harmonic of the magnetic and electric Mie coefficients is unity for both forward and backward scattering cases. Moreover, for the forward scattering condition, the magnetic and electric Mie coefficients pick up the same phase at the resonant frequency, while for the backward scattering scenario, there is a π relative phase difference between them. Fig. 5 summarizes the scatterer's performance for both the MD and ED resonant frequencies. For the MD excitation, Fig. 5(a) and Fig. 5(e) show the in-plane component of the electric field in the near-field and far-field regions, respectively. It can be observed that the in-phase interaction of the MD and ED modes leads to suppressed backward scattering, as predicted

by our earlier discussions. On the contrary, the in-plane component of the electric field plotted in the near-field and far-field regions for the MD excitation in Fig. 5(b) and Fig. 5(f), respectively, illustrates that the out-of-phase interference of the excited MD and ED modes in the scatterer effectively vanquishes forward scattering.

Moreover, for the ED excitation, Fig. 5(c) and Fig. 5(g) illustrate the in-plane electric field in the near-field and far-field regions, respectively. The results demonstrate that the proper choice of the modulation function can effectively excite the MD and ED modes accompanied by the same moments within the particle. Consequently, the in-phase interference between these excited modes leads to a perfect suppression of backward scattering. Finally, for the ED excitation, the in-plane component of the electric field is depicted in Fig. 5(d) and Fig. 5(h) for the near-field and far-field regions, respectively. As observed, the out-of-phase interaction of the excited MD and ED modes in the particle results in total backward scattering. These findings highlight the significant impact of manipulating the eigenvalues of the dynamic medium to achieve selective suppression of either backward or forward scattering, depending on the desired functionality.

In summary, our study presents a pioneering analytical solution for scattering from a particle composed of a material with time-varying permittivity, revealing a dynamic T-matrix with controllable elements. By skillfully selecting modulation functions, we harnessed this capability to dynamically satisfy the Kerker conditions within the scatterer without the need to alter its geometrical parameters or introduce magnetic properties. This groundbreaking approach offers a powerful platform for precisely controlling the scattering characteristics of individual meta-atoms, paving the way for designing more complex and sophisticated time-varying structures. The ability to control the scattering characteristics in such a precise manner opens up promising opportunities for state-of-the-art light-matter interactions and advanced wavefront engineering, enabling the design of next-generation optical devices and systems with enhanced performance and capabilities. The findings of this research shed light on the future of light-matter interactions, providing beneficial insights and opportunities for advancing optical technologies and exploring exotic applications in the realm of dynamic materials.

SUPPLEMENTARY MATERIAL

Detailed description of the analytical approach, and comments on the genetic algorithm optimization.

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